C Programming

Course Title: C Programming **Full Marks**: 60 + 20 + 20 Pass Marks: 24 + 8 + 8

Course No: BIT102 **Credit Hrs**: 3

Nature of the Course: Theory + Lab

Semester: I

Course Description: This course familiarizes students with basic principles of programming. It introduces structured programming paradigm using a high level language called C. It covers the concept of problem solving techniques, program design, and basic elements of C along with the detailed concept of operators, statements, arrays, functions, pointers, structures and file handling.

Course Objective: The main objective of this course is to acquaint the students with good program design through structured programming paradigm for developing programs for specific tasks using C Programming Language as well as to present the syntax and semantics of the "C" language.

Course Contents:

Unit 1: Introduction (3 Hrs.)

History and advantages of C; Problems analysis, algorithm and flowchart; Structure of a C Program; Writing, compiling, debugging, executing and testing a C Program in windows and Unix/Unix like environment

Unit 2: Elements of C (3 Hrs.) C Tokens; C Character Sets; Data types, Constants and Variables; Expression, statements and comments; Escape sequences and Delimiters

Unit 3: Input/output function (2 Hrs.) Conversion Specifiers; I/O functions; Formatted I/O
Unit 4: Operators and Expression (4 Hrs.) Arithmetic operators; Relational operators; Logical operators; Assignment operators; Type conversion in assignment; Increment and decrement operators; Ternary operator; Bitwise operator; Other operators (comma, sizeof); Expression evaluation; Operator precedence and associativity

Unit 5: Control Structures (8 Hrs.) Introduction to selection and iterative statements; GOTO and labels; Selection statements: if, if..else, if..else if ladder, nested if, switch case; Conditional operator; Iterative statement: For Loop, While Loop, Do while Loop, Nested Loop; The odd loop; Controlling the loop execution – break and continue

Unit 5: Arrays and Strings (5 Hrs.) Introduction to Arrays; Initializing Arrays; The meaning of array indexing; One dimensional and Multidimensional Arrays; String and Basic functions dedicated to string manipulation 3

Unit 6: Functions (6 Hrs.) Introduction and types of functions; Declaring, Defining and calling functions; Arguments and Return Statement; Recursive functions; Function call by value and reference; Variables' scope, local variables and function parameters; Arrays as function parameter; Void as a parameter; Parameterizing the main function; External function and variables; Header files; Static variables; Register Variables

Unit 7: The C Preprocessor (2 Hrs.) Features of C Preprocessor; Macro Expansion; Macros with Arguments; Macros versus Functions; File Inclusion; Conditional Compilation; #if and #elif Directives; #undef Directive; #pragma Directive; The Build Process; Preprocessing; Compiling; Assembling; Linking; Loading

Unit 8: Pointers (5 Hrs.) Introduction of Pointers, declaration and initialization of pointer variables; An address, a reference, a deference and the size of operator; Pointer to nothing (NULL); Pointer assignment; Pointer Arithmetic; Pointer as argument and Pointer as return values; Pointers vs. arrays; Dynamic memory allocation

Unit 9: Structure and Unions (5 Hrs.) Definition of Structure; Array of structures; Passing structure and array of structure to function; Pointers to structures and arrays of structures; Self-referential structures; Typedef; Table Lookup; Unions

Unit 10: File Handling (2 Hrs.) Files vs. streams; Header files needed for stream operations; Opening and closing a stream, open modes, errno variable; Reading and writing to/from a stream; Predefined streams: stdin, stdout and stderr; Stream manipulation: fgetc(), fputc(), fgets() and fputs() functions; Raw input/output: fread() and fwrite() functions; Random access to files

Laboratory Works: Laboratory work emphasizes the verification of programming concepts learned in class. Therefore, each unit should include sufficient practical lab exercise.

Text / Reference Books: 1. Let Us C, Yashavant P. Kanetkar 2. Brian Kernighan and Dennis Ritchie, The C Programming Language 3. Byron Gottfried, Programming with C, McGraw Hill Education